

Section 9.2

Exercise (9.2.1). Let $f(x) \in F[x]$ be a polynomial of degree $n > 1$ and let bars denote passage to the quotient $F[x]/(f(x))$. Prove that for each $\overline{g(x)}$ there is a unique polynomial $g_0(x)$ of degree $\leq n - 1$ such that $\overline{g(x)} = \overline{g_0(x)}$.

Proof: Let F be a field and let x be indeterminate over F also let $f(x) \in F[x]$ be a polynomial of degree $n > 1$. Suppose $g(x) \in F[x]$, since F is a field we know that $F[x]$ is a Euclidean Domain, applying the division algorithm we get that there are unique $q(x)$ and $r(x)$ in $F[x]$ such that,

$$g(x) = q(x)f(x) + r(x).$$

with $r(x) = 0$ or degree $\deg(r(x)) < \deg(f(x)) = n$. Note that in the quotient $F[x]/(f(x))$, $\overline{g(x)} = \overline{r(x)}$ and $r(x)$ is unique with degree $\leq n - 1$. $\square$

Exercise (9.2.2). Let F be a finite field of order q and let $f(x)$ be a polynomial in $F(x)$ of degree $n \geq 1$. Prove that $F[x]/(f(x))$ has q^n elements.

Proof: Suppose F is a finite field of order q and let $f(x)$ be a polynomial in $F(x)$ of degree $n \geq 1$. By the previous result we know that the elements $\overline{1}, \overline{x}, \dots, \overline{x^{n-1}}$ are a basis of the vector space $F[x]/(f(x))$ over F - in particular $F[x]/(f(x))$ is dimension n over F . Since F is order q and $F[x]/(f(x))$ is dimension n over F it must follow that $F[x]/(f(x))$ has q^n elements. $\square$

Exercise (9.2.3). Let $f(x)$ be a polynomial in $F[x]$. Prove that $F[x]/(f(x))$ is a field if and only if $f(x)$ is irreducible. [Use Proposition 7, Section 8.2]

Proof: Let F be a field with x indeterminate over F and $f(x) \in F[x]$. Since F is a field we know that $F[x]$ is a Euclidean Domain and is also therefore a Principal Ideal Domain. Suppose $F[x]/(f(x))$ is a field, therefore by Proposition 12, Section 7.4 ($f(x)$) must

have been a maximal ideal in $F[x]$. Since $F[x]$ is commutative and $(f(x))$ is maximal, by Corollary 14, Section 7.4 $(f(x))$ is prime and therefore irreducible.

Suppose $(f(x))$ is irreducible, it is therefore prime and also by Proposition 7, Section 8.2 $(f(x))$ is also maximal in $F[x]$. Again by Proposition 12, Section 7.4 $F[x]/(f(x))$ is a field. $\square$

Exercise (9.2.7). Determine all the ideals of the ring $\mathbb{Z}/(2, x^3 + 1)$.

Proof: $\square$

Section 9.3

Exercise (9.3.1). Let R be an integral domain with quotient field F and let $p(x)$ be a monic polynomial in $R[x]$. Assume that $p(x) = a(x)b(x)$ where $a(x)$ and $b(x)$ are monic polynomials in $F[x]$ of smaller degree than $p(x)$. Prove that if $a(x) \notin R[x]$ then R is not a UFD. Deduce that $\mathbb{Z}[2\sqrt{2}]$ is not a UFD.

Proof: Let R be an integral domain with quotient field F and let $p(x)$ be a monic polynomial in $R[x]$. Also let $p(x) = a(x)b(x)$ where $a(x)$ and $b(x)$ are monic polynomials in $F[x]$ of smaller degree than $p(x)$. Suppose $a(x) \notin R[x]$ and R is a UFD. Since R is a UFD, and $p(x)$ is reducible in $F[x]$, i.e. $p(x) = a(x)b(x)$ in $F[x]$, by Gauss' Lemma there are $r, s \in F$ such that $ra(x), sb(x) \in R[x]$ where $p(x) = ra(x)sb(x)$ is a factorization in $R[x]$. Consider that the leading coefficient of $rsa(x)b(x)$ is 1 and $r, s \in R$, since $a(x)$ and $b(x)$ are monic polynomials, therefore we conclude that $rs = 1$ so finally we get that $rsa(x) = a(x) \in R[x]$ a contradiction.

Consider $x^2 - 2 \in \mathbb{Z}[2\sqrt{2}]$ and note that $x^2 - 2 = (x + \sqrt{2})(x - \sqrt{2})$ in the field of fractions $\mathbb{Q}[2\sqrt{2}]$ since $\frac{1}{2}(2\sqrt{2}) = \sqrt{2} \in \mathbb{Q}[2\sqrt{2}]$, yet $(x + \sqrt{2}), (x - \sqrt{2}) \notin \mathbb{Z}[2\sqrt{2}]$ so by the previous result $\mathbb{Z}[2\sqrt{2}]$ is not a UFD. $\square$

Exercise (9.3.3). Let F be a field. Prove that the set R of polynomials in $F[x]$ whose coefficient of x is equal to 0 is a subring of $F[x]$ and that R is not a UFD.

Proof: Let F be a field. Suppose R is the set of polynomials in $F[x]$ whose coefficient of x is equal to zero. Let $p(x), q(x) \in R$ with coefficients $a_i, b_i \in F$ respectively, and $\deg(p) = m, \deg(q) = n$. Note that $p(x) - q(x)$ will still have a coefficient of zero on the x term. Consider $p(x)q(x)$, and recall that the coefficient on the x term will be given by the following sum, $a_0b_1 + a_1b_0 = 0$ thus R is a subring of $F[x]$.

Since there are no polynomials with degree 1 in R , reducing $x^2 = a(x)b(x)$ where both $a(x), b(x)$ are nonzero is impossible so x^2 is irreducible in R . Similarly we find that x^3 is irreducible in R . Yet we find that x^6 has two distinct factorizations into irreducibles, namely $x^6 = x^2x^2x^2$ and $x^6 = x^3x^3$, thus R is not a UFD. $\square$

Section 9.4

Exercise (9.4.1). Determine whether the following polynomials are irreducible in the rings indicated. For those that are reducible, determine their factorization into irreducibles. The notation $\mathbb{F}_p$ denotes the finite field $\mathbb{Z}/p\mathbb{Z}$, p a prime.

- (a) $x^2 + x + 1$ in $\mathbb{F}_2[x]$.

Proof: Recall Proposition 10, which states a polynomial of degree two or three over field F is reducible if and only if it has a root in F . Note that $(0)^2 + (0) + 1 = 1$ and $(1)^2 + (1) + 1 = 1$ so by the converse of the forward direction of Proposition 10, $x^2 + x + 1$ is irreducible. Also this is just Example (4) on p.308. $\square$

- (b) $x^3 + x + 1$ in $\mathbb{F}_3[x]$.

Proof: Again we can check if there are any roots and apply Proposition 10. Note that, $(1)^3 + (1) + 1 = 0$. So we get that $(x - 1)$, or more accurately $(x + 2)$ is a factor of $x^3 + x + 1$ in $\mathbb{F}_3[x]$. $\square$

- (c) $x^4 + 1$ in $\mathbb{F}_5[x]$.

Proof: Note that in $\mathbb{F}_5[x]$, $x^4 + 1 = x^4 - 4$. A difference of squares, which can still be factored in $\mathbb{F}_5[x]$, $x^4 + 1 = x^4 - 4 = (x^2 + 2)(x^2 - 2) = (x^2 + 2)(x^2 + 3)$ in $\mathbb{F}_5[x]$. Thus $x^4 + 1$ is reducible in $\mathbb{F}_5[x]$ with factors $(x^2 + 2)$ and $(x^2 + 3)$. $\square$

(d) $p(x) = x^4 + 10x^2 + 1$ in $\mathbb{Z}$.

Proof: Let $g(x) = p(3x + 1) = (3x + 1)^4 + 10(3x + 1)^2 + 1 = 81x^4 + 108x^3 + 144x^2 + 72x + 12$. Applying Eisenstien's Criterion on $g(x)$ with $p = 3$ we get that $g(x)$ is irreducible. Therefore it follows that $p(x)$ is also irreducible since any factorization of $p(x)$, by substitution would give a factorization for $g(x)$. ???
Questionable $g(x)$ is not a monic polynomial.

$\square$

Exercise (9.4.2). Prove that the following polynomials are irreducible in $\mathbb{Z}[x]$:

(a) $x^4 - 4x^3 + 6$

Proof: Applying Eisenstien's Criterion with $p = 2$ we get that $x^4 - 4x^3 + 6$ is irreducible in $\mathbb{Z}[x]$. $\square$

(b) $x^6 + 30x^5 - 15x^3 + 6x - 120$

Proof: Applying Eisenstien's Criterion with $p = 3$ we get that $x^6 + 30x^5 - 15x^3 + 6x - 120$ is irreducible in $\mathbb{Z}[x]$. $\square$

(c) $p(x) = x^4 + 4x^3 + 6x^2 + 2x + 1$

Proof: Consider $g(x) = p(x - 1) = (x - 1)^4 + 4(x - 1)^3 + 6(x - 1)^2 + 2(x - 1) + 1 = x^4 - 2x + 2$. Applying Eisenstien's Criterion with $p = 2$ we find that $g(x)$ is irreducible and therefore $p(x)$ is also irreducible in $\mathbb{Z}[x]$. $\square$

(d) $\frac{(x+2)^p - 2^p}{x}$ where p is an odd prime.

Proof: Expanding our polynomial using the Binomial Theorem we get,

$$\frac{(x+2)^p - 2^p}{x} = \frac{\left(\sum_{k=0}^p \binom{p}{k} x^{p-k} 2^k\right) - 2^p}{x} = \frac{\left(\sum_{k=0}^{p-1} \binom{p}{k} x^{p-k} 2^k\right)}{x} = \sum_{k=0}^{p-1} \binom{p}{k} x^{p-k-1} 2^k.$$

Note that the degree 0 term in this polynomial is $2^{p-1} \binom{p}{p-1} = 2^{p-1}(p)$ which since p is odd, p^2 does not divide p . Since $\binom{p}{k}$ is divisible by p for $k > 0$, and our polynomial is monic since $\binom{p}{0} 2^0 = 1$, we conclude by Eisenstien's Criterion with prime p that $\frac{(x+2)^p - 2^p}{x}$ is irreducible.

□

Exercise (9.4.5). Find all monic irreducible polynomials of degree ≤ 3 in $\mathbb{F}_2[x]$, and the same in $\mathbb{F}_3[x]$.

Solution: To find all monic irreducible polynomials we will use Proposition 10 as we did in problem 1 and check if all monic polynomials of degree ≤ 3 in $\mathbb{F}_2[x]$ have roots,

Monic Polynomial	Roots	Irreducible?
$x^3 + x^2 + x + 1$	1	reducible
$x^3 + x^2 + x$	0	reducible
$x^3 + x^2 + 1$	none	irreducible
$x^3 + x + 1$	none	irreducible
$x^2 + x + 1$	none	irreducible
$x^3 + x^2$	0	reducible
$x^3 + x$	0	reducible
$x^3 + 1$	1	reducible
$x^2 + x$	0	reducible
$x^2 + 1$	1	reducible
x^3	0	reducible
x^2	0	reducible

Therefore we get that $x^3 + x^2 + 1$, $x^3 + x + 1$, $x^2 + x + 1$, $x + 1$, x , and 1 are irreducibles in $\mathbb{F}_2[x]$. $\square$

Exercise (9.4.6). Construct fields for which the following orders: (a)9, (b)49, (c)8, and (d)81.

(a) A field of order 9.

Proof: Consider $F = \mathbb{F}_3[x]$ a field of order 3. Let $f(x) = x^2 + 1$ an irreducible polynomial of order 2 in F . By Exercise 2 of Section 9.2 we know that $F/(f(x))$ is order 9. $\square$

(b) A field of order 49.

Proof: Consider $F = \mathbb{F}_7[x]$ a field of order 7. Let $f(x) = x^2 + 1$ an irreducible polynomial of order 2 in F . By Exercise 2 of Section 9.2 we know that $F/(f(x))$ is order 49. $\square$

(c) A field of order 8.

Proof: Consider $F = \mathbb{F}_2[x]$ a field of order 2. Let $f(x) = x^3 + x + 1$ an irreducible polynomial of order 3 in F . By Exercise 2 of Section 9.2 we know that $F/(f(x))$ is order 8. $\square$

(d) A field of order 81.

Proof: Consider $F = \mathbb{F}_3[x]$ a field of order 3. Let $f(x) = x^4 + x + 2$, showing that $f(x)$ is irreducible in F implies that $F/(f(x))$ is order $3^4 = 81$ as we have done in previous exercises. Clearly $f(x) = x^4 + x + 2$ has no linear factors as it has no roots in $\mathbb{F}_3$, $f(0) = 1$, $f(1) = 1$, and $f(2) = 2$. Suppose for the sake of contradiction that $f(x)$ has quadratic factors

$$f(x) = (x^2 + ax + b)(x^2 + cx + d) = x^4 + (a+c)x^3 + (ac+b+d)x^2 + (ad+bc)x + bd.$$

Therefore we get that $a + c = 0$, $ac + b + d = 0$, $ad + bc = 1$, and $bd = 2$.

Since $c = -a$, by substitution we get $ad + bc = a(d - b) = 1$. So $b \neq d$ and

since $bd = 2$, either $b = 2$ and $d = 1$ or vice versa. Let $b = 2$ and $d = 1$ then $a(d - b) = 1$ gives that $a = -1$ and therefore $c = 1$, however we get that $-1 = (-1)(1) + 2 + 1 = ac + b + d = 0$ a contradiction. Let $b = 1$ and $d = 2$ then $a(d - b) = 1$ gives that $a = 1$ and therefore $c = -1$, however $-1 = (1)(-1) + 1 + 2 = ac + b + d = 0$ so we arrive at another contradiction. Therefore $f(x)$ is irreducible in F and thus $F/(f(x))$ is order $3^4 = 81$. $\square$

Exercise (9.4.18). Show that $6x^5 + 14x^3 - 21x + 35$ and $18x^5 - 30x^2 + 120x + 360$ are irreducible over $\mathbb{Q}[x]$.

Proof: Applying Eisenstien's Criterion with prime $p = 7$ to show that $6x^5 + 14x^3 - 21x + 35$ is irreducible over $\mathbb{Z}[x]$ and also applying Eisenstein's Criterion with prime $p = 5$ to show that $18x^5 - 30x^2 + 120x + 360$ is irreducible over $\mathbb{Z}[x]$. By the contrapositive of Gauss's Lemma we can conclude that both polynomials are irreducible over $\mathbb{Q}[x]$. $\square$

Section 9.5

Exercise (9.5.1). Let F be a field and let $f(x)$ be a nonconstant polynomial in $F[x]$. Describe the nilradical of $F[x]/(f(x))$ in terms of the factorization of $f(x)$.

Proof: Suppose F is a field and let $f(x)$ be a nonconstant polynomial in $F[x]$. Let $f(x) = f_1(x)^{n_1} f_2(x)^{n_2} \dots f_k(x)^{k_1}$ be its factorization into distinct irreducibles. Recall that the nilradical of $F[x]/(f(x))$ is the set of nilpotent elements of $F[x]/(f(x))$. If a polynomial $g(x)^n$ in $F[x]$ is some multiple $f(x)$, i.e $f(x) \mid g(x)^n$, then $\overline{g(x)}$ will be nilpotent in $F[x]/(f(x))$ and contained in the nilradical of $F[x]/(f(x)) \dots$

Not sure where to go from here... The nilradical of $F[x]/(f(x))$ is the set of nilpotent elements of $F[x]/(f(x))$. The elements of $F[x]/(f(x))$ look like the remainders... $\square$

Exercise (9.5.3). Let p be an odd prime in $\mathbb{Z}$ and let n be a positive integer. Prove that $x^n - p$ is irreducible over $\mathbb{Z}[i]$.

Proof: Suppose p is an odd prime in $\mathbb{Z}$ and let n be a positive integer. Since p is an odd prime in $\mathbb{Z}$ it is either $p \cong 3 \pmod{4}$ or $p \cong 1 \pmod{4}$. By Proposition 18 (2.b) we know that if $p \cong 3 \pmod{4}$ then p is irreducible in $\mathbb{Z}[i]$. In which case (p) is a prime ideal in $\mathbb{Z}[i]$ which satisfies Eisenstein's Criterion since $p \notin (p)^2$. From Proposition 18 (2.c) we get that if $p \cong 1 \pmod{4}$ then $p = (a + bi)(a - bi)$ where $(a + bi)$ and $(a - bi)$ are irreducible in $\mathbb{Z}[i]$. In which case $(a + bi)$ is a prime ideal in $\mathbb{Z}[i]$ which satisfies Eisenstein's Criterion since $p \notin (a + bi)^2$. So $x^n - p$ is irreducible over $\mathbb{Z}[i]$. $\square$